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Project Report

(PETV140)

Topic: **CINEWAVE – MOVIE TICKET BOOKING
SYSTEM**

Submitted to:

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Project Links:

GitHub Repository:

<https://github.com/Rishav-Thakur1314/movie-ticket-booking-system>

Deployment Link:

<https://movie-ticket-booking-system-nvc98cf5p-moviebooking.vercel.app/>

2. Abstract

The “**CineWave – Movie Ticket Booking System**” is a full-stack web application designed to simplify the process of booking movie tickets online. The system provides users with a secure and user-friendly platform to browse available movies, select preferred seats, and manage bookings efficiently. It eliminates the inconvenience of traditional ticket booking by offering an online solution that is accessible anytime and from any device.

Objectives

- Develop an online platform for seamless movie ticket booking.
- Provide secure user authentication and booking management.
- Offer an intuitive and responsive user interface.
- Ensure efficient storage and retrieval of booking information.

Technologies Used

Technology	Purpose
React.js	Frontend Development
Vite	Build Tool
Tailwind CSS	User Interface Styling
Supabase	Backend Services & Authentication
PostgreSQL	Database Management
React Context API	State Management
Vercel	Application Deployment

Key Features

- User Registration and Secure Login
- Browse Available Movies
- View Movie Details
- Seat Selection Interface
- Online Ticket Booking
- Booking History Management
- Responsive Design for Multiple Devices
- Secure Database with User-Specific Access

Outcome

The project successfully demonstrates the implementation of a modern web-based movie ticket booking system using current full-stack development technologies. It provides a scalable, secure, and responsive solution while applying software engineering principles such as component-based architecture, client-server communication, authentication, database management, and modular application design.

3. Introduction

3.1 Project Overview

The “**CineWave – Movie Ticket Booking System**” is a web-based application developed to simplify and digitize the movie ticket booking process. The application allows users to browse available movies, create an account, log in securely, select preferred seats, and book tickets through an intuitive interface. By integrating modern frontend and backend technologies, the system provides a seamless booking experience while ensuring secure storage and management of user and booking data.

The project is developed using **React.js** with **Vite** for the frontend, **Tailwind CSS** for responsive user interface design, and **Supabase** as the backend service for authentication and database management. The database is built on **PostgreSQL**, ensuring reliable data storage and efficient query execution. The application is deployed on **Vercel**, making it accessible through any modern web browser.

The system follows a modular architecture where each functionality, such as authentication, movie listing, and booking management, is implemented as an independent module. This improves code maintainability, scalability, and reusability while providing a smooth user experience.

3.2 Problem Statement

Traditional movie ticket booking methods often involve manual processes or limited digital systems, leading to several challenges for both customers and theatre management.

Problems in Traditional Booking System

- Long waiting queues at ticket counters.
- Difficulty in checking seat availability in real time.
- Manual maintenance of booking records.
- Higher chances of booking errors.
- Limited accessibility and inconvenience for users.

- Increased workload for theatre staff.

These challenges create the need for a modern online booking platform that allows users to reserve tickets quickly and securely from anywhere.

3.3 Objectives

The primary objectives of the “**CineWave Movie Ticket Booking System**” are:

- To develop a secure and user-friendly online movie ticket booking platform.
 - To provide authenticated access for registered users.
 - To allow users to browse available movies and view their details.
 - To enable easy seat selection and ticket booking.
 - To maintain booking information securely in a cloud database.
 - To create a responsive application that works across different devices.
 - To implement a scalable architecture using modern web development technologies.
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3.4 Existing System

In many traditional movie theatres, ticket booking is performed either manually or through basic online systems with limited functionality. These methods often fail to provide a smooth and efficient booking experience.

Limitations of the Existing System

- Manual ticket booking requires physical presence.
 - Long waiting time during weekends and holidays.
 - Lack of real-time seat availability.
 - Poor user experience due to outdated interfaces.
 - Limited accessibility from multiple devices.
 - Difficult record management and maintenance.
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3.5 Proposed System

The proposed “**CineWave Movie Ticket Booking System**” addresses the limitations of traditional booking methods by providing a secure and responsive web application.

Features of the Proposed System

- Secure user registration and login.
- Browse available movies with detailed information.
- Interactive seat booking interface.
- Store booking records securely in PostgreSQL.
- Responsive design for desktop and mobile devices.
- Cloud-based backend using Supabase.
- Fast deployment and accessibility through Vercel.

The modular architecture allows future enhancements such as online payments, theatre management, movie reviews, and ticket cancellatio

3.6 Scope of the Project

The “**CineWave – Movie Ticket Booking System**” is designed to provide a simple, secure, and efficient platform for booking movie tickets online. The project focuses on improving the user experience while demonstrating the implementation of modern full-stack web development technologies.

Current Scope

- Online user registration and login.
- Browse available movies.
- View movie details.
- Select preferred seats.
- Book movie tickets online.
- Store booking details securely in the database.
- View booking history.

- Responsive interface for desktop and mobile devices.

Future Scope

- Online payment gateway integration.
- Ticket cancellation and refund system.
- Movie ratings and reviews.
- Admin dashboard for theatre management.
- Email and SMS booking confirmation.
- QR Code-based ticket generation.
- Real-time seat availability updates.
- Multiple theatre and city support.

3.7 Technology Stack

The project utilizes modern web development technologies to build a scalable and responsive application.

Technology	Purpose
React.js	Frontend development using reusable components
Vite	Fast build tool and development server
JavaScript (ES6)	Application logic
Tailwind CSS	Responsive UI styling
React Context API	Global state management
React Router DOM	Client-side routing
Supabase	Backend services and authentication
PostgreSQL	Database management
Vercel	Deployment platform

Technology

Purpose

Advantages of the Selected Technologies

- Fast application performance.
 - Component-based architecture.
 - Responsive user interface.
 - Secure cloud database.
 - Easy scalability and maintenance.
 - Simple deployment process.
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3.8 Software Requirements

Development Software

Software

Purpose

Visual Studio Code	Code Editor
Node.js	JavaScript Runtime
npm	Package Manager
Git	Version Control
Supabase	Backend & Database
Vercel	Cloud Deployment
Google Chrome	Testing Browser

Programming Languages

- JavaScript
- HTML5
- CSS3

- SQL

3.9 Hardware Requirements

Minimum Requirements

Component Specification

Processor	Intel Core i3 / Ryzen 3
RAM	4 GB
Storage	20 GB Free Space
Internet	Broadband Connection
Browser	Chrome / Edge / Firefox

Recommended Requirements

Component Specification

Processor	Intel Core i5 / Ryzen 5 or Above
RAM	8 GB or Above
Storage	SSD (50 GB Free Space)
Internet	High-Speed Broadband
Browser	Latest Chrome or Edge

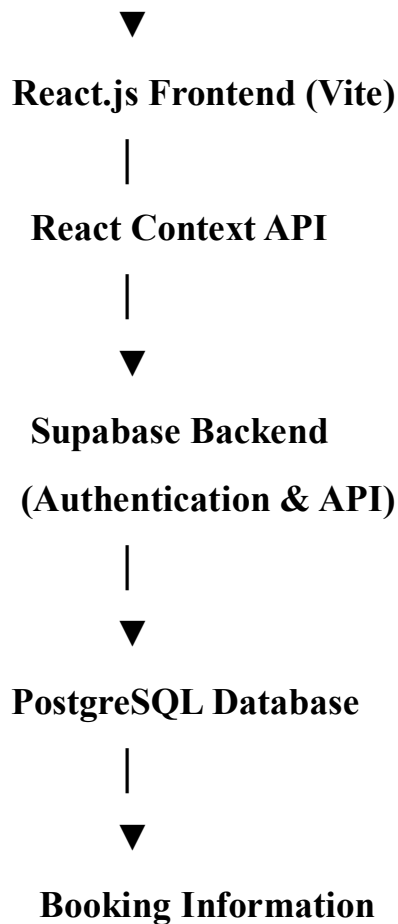
3.10 System Architecture

The “CineWave Movie Ticket Booking System” follows a **Client–Server Architecture**, where the frontend communicates with the backend services provided by Supabase.

Architecture Workflow

User

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Working Flow

1. User opens the website.
2. User registers or logs in.
3. Available movies are fetched from the database.
4. User selects a movie.
5. User books the preferred seat.
6. Booking details are stored in PostgreSQL.
7. User can view their booking history.

3.11 Database Design

The application uses **PostgreSQL** through Supabase to store and manage application data.

Main Database Tables

Table Name	Purpose
Movies	Stores movie information
Bookings	Stores user booking details

Movies Table

Stores:

- Movie ID
- Movie Name
- Description
- Genre
- Duration
- Poster Image
- Show Timing

Bookings Table

Stores:

- Booking ID
- User ID
- Movie ID
- Seat Number
- Booking Date
- Booking Status

Database Features

- UUID Primary Keys
- Foreign Key Relationships
- Secure User Authentication
- Row Level Security (RLS)
- Indexed Queries for Better Performance

3.12 Project Modules

The “CineWave” application is divided into multiple functional modules to improve maintainability and scalability.

1. Authentication Module

Functions:

- User Registration
- User Login
- Secure Authentication
- Session Management

2. Movie Management Module

Functions:

- Display Movies
- View Movie Details
- Browse Available Shows

3. Booking Module

Functions:

- Seat Selection
- Ticket Booking
- Store Booking Details
- Prevent Duplicate Bookings

4. User Dashboard Module

Functions:

- View Booking History

- Manage User Account
- Access Previous Bookings

5. Database Module

Functions:

- Store Movie Information
- Store Booking Records
- Manage User Data
- Perform CRUD Operations

4. Algorithm

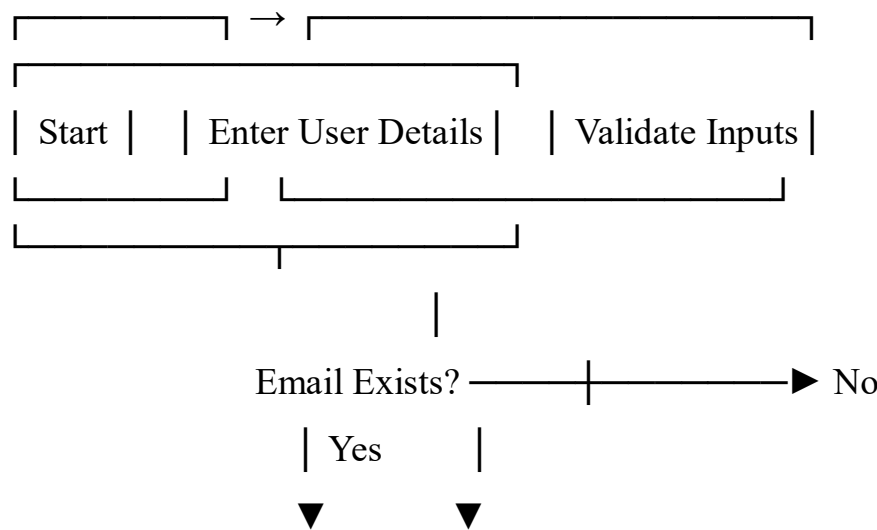
The “CineWave Movie Ticket Booking System” follows a modular approach where each major functionality is implemented using a separate algorithm. The following algorithms describe the workflow of the application's core modules.

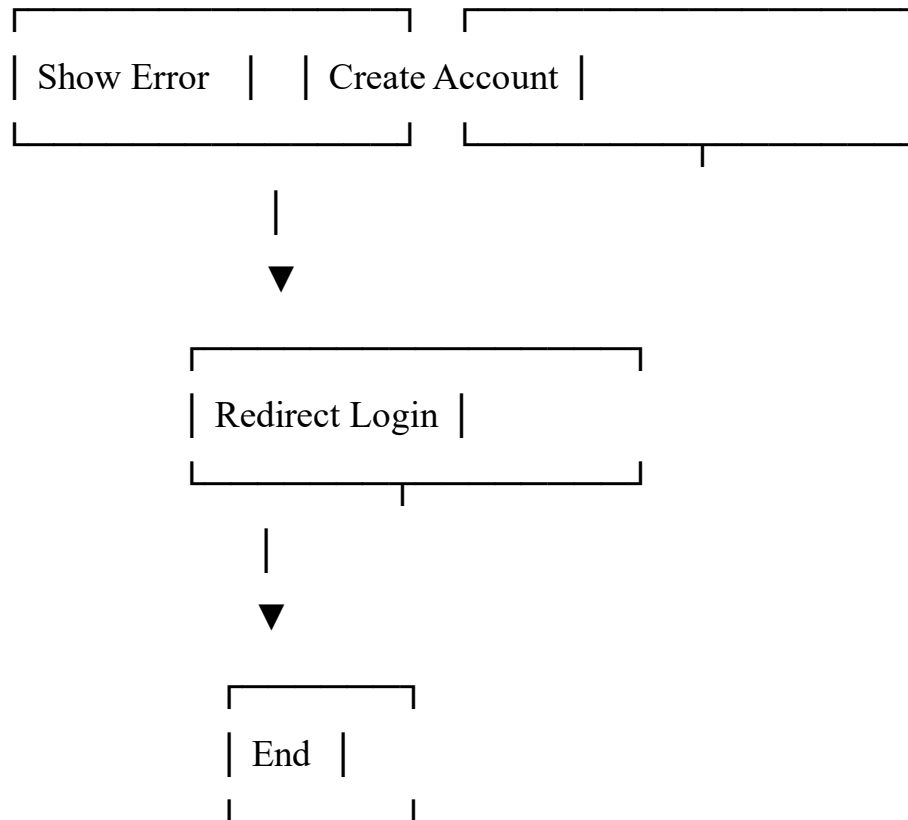
4.1 User Registration Algorithm

Objective

To create a new user account using Supabase Authentication.

Flowchart





Algorithm Steps

1. User enters Name, Email, and Password.
2. Validate all input fields.
3. Check whether the email is already registered.
4. If the email exists, display an error message.
5. Otherwise, create a new account.
6. Store user credentials securely.
7. Display a success message.
8. Redirect the user to the Login page.

Time Complexity $O(1)$

Space Complexity $O(1)$

DSA Concepts Used

- String Validation
- Authentication

- Database Lookup

4.2 User Login Algorithm

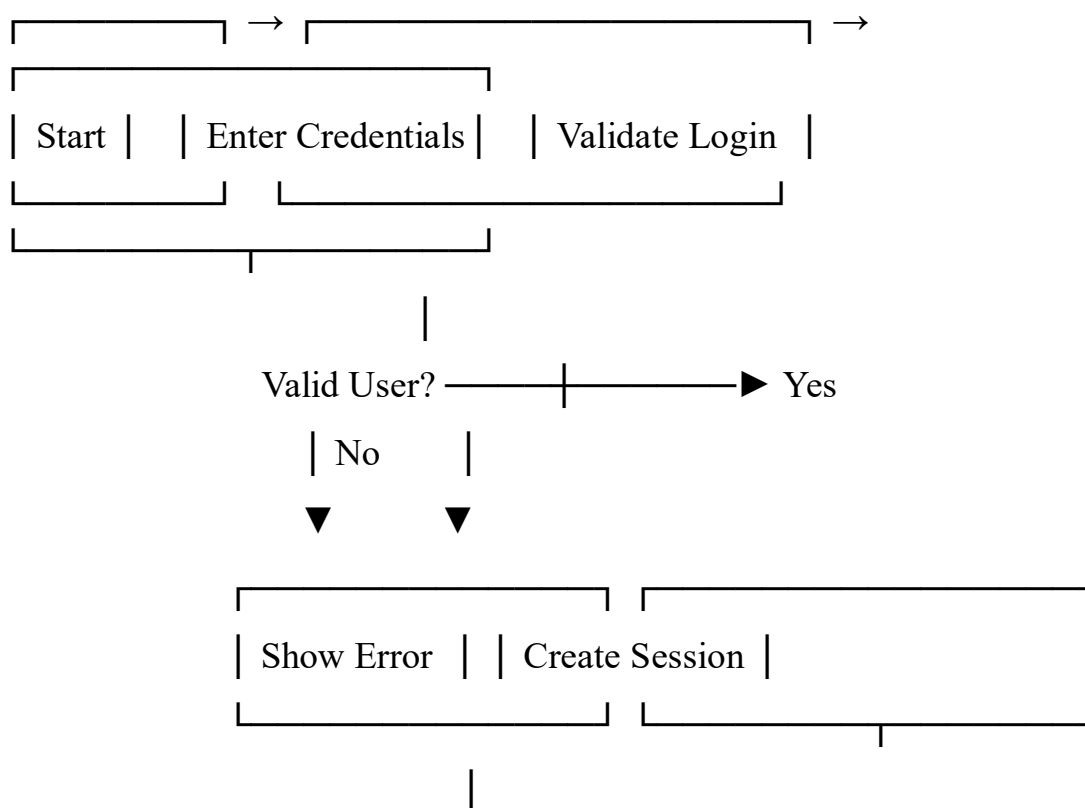
Objective

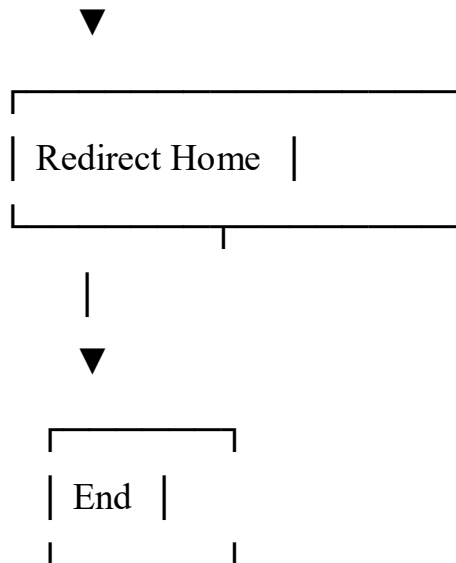
To authenticate users and grant secure access.

Algorithm Steps

1. User enters Email and Password.
2. Validate credentials.
3. Send authentication request.
4. If authentication fails, display an error.
5. If successful, create a session.
6. Redirect to Home Page.

Flowchart





Time Complexity $O(1)$

Space Complexity $O(1)$

DSA Concepts Used

- String Matching
- Authentication
- Session Management

4.3 Movie Fetching Algorithm

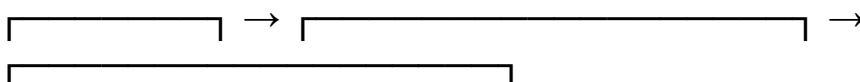
Objective

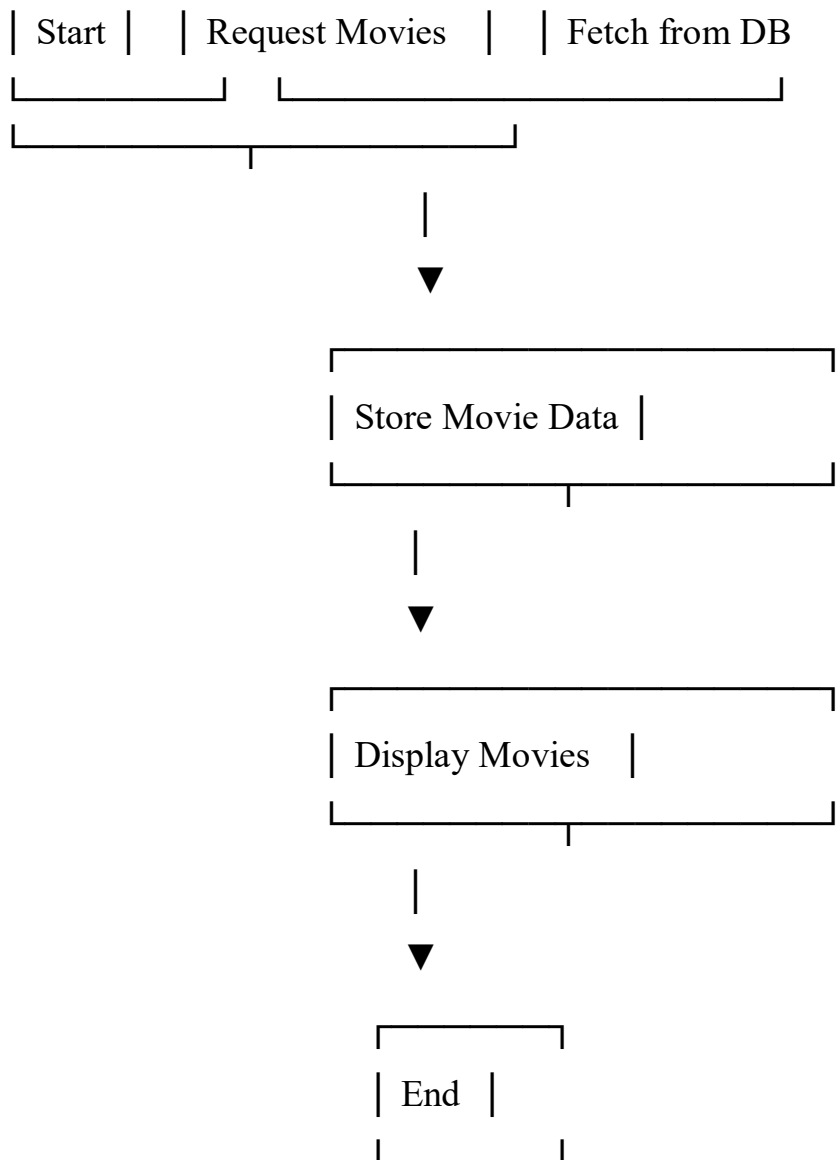
To retrieve available movies from the database.

Algorithm Steps

1. Open Home Page.
2. Request movie data from Supabase.
3. Receive movie records.
4. Store the records.
5. Display movie cards.

Flowchart





Time Complexity $O(n)$

Space Complexity $O(n)$

DSA Concepts Used

- Array Traversal
- Database Query
- Data Rendering

4.4 Seat Selection Algorithm

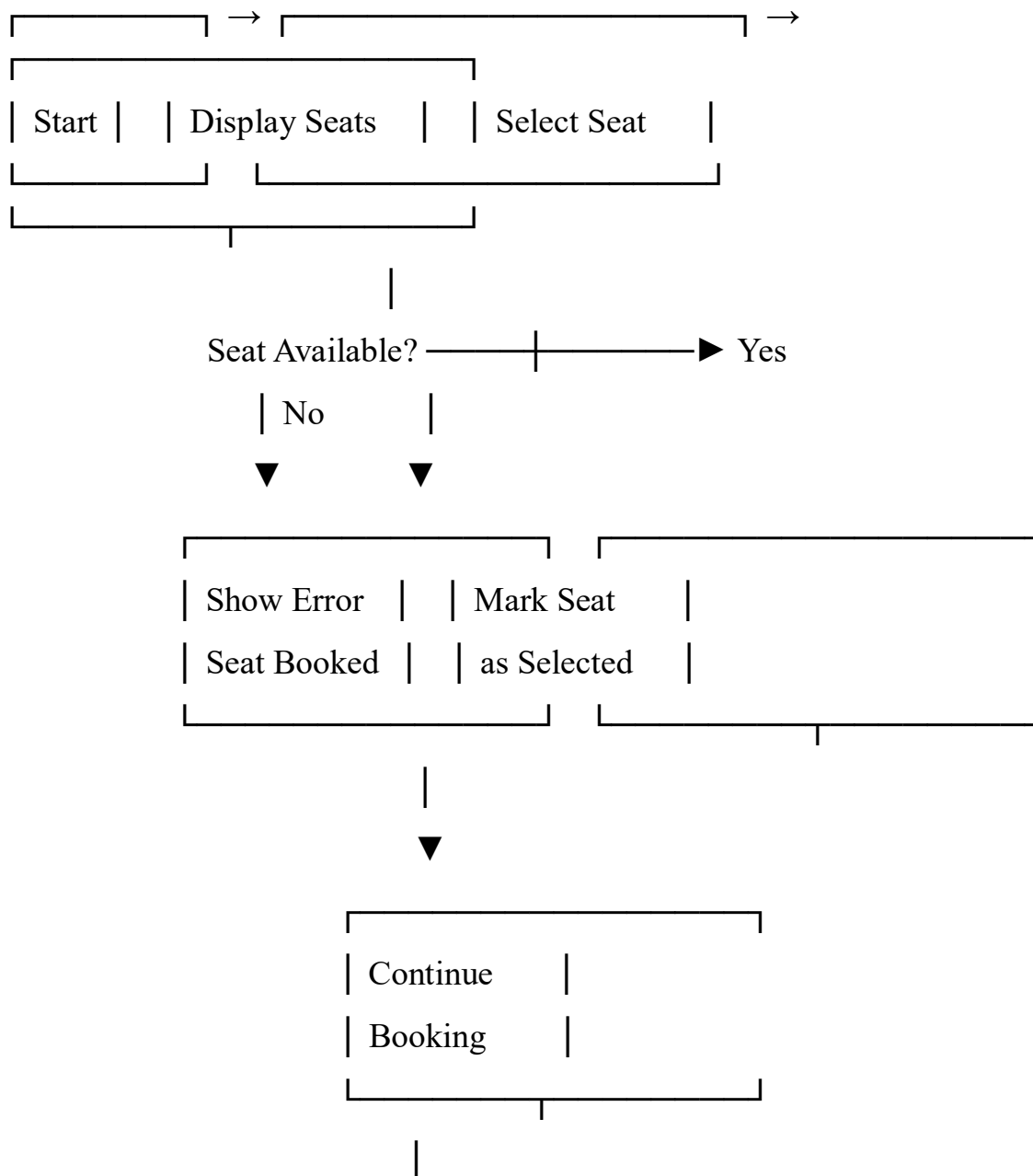
Objective

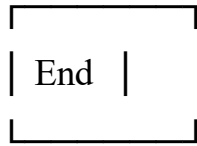
To allow users to select only available seats for booking.

Algorithm Steps

1. Display seat layout.
2. User selects a seat.
3. Verify seat availability.
4. If already booked, show an error.
5. Otherwise, mark the seat as selected.
6. Continue to booking confirmation.

Flowchart





Time Complexity $O(n)$

Space Complexity $O(n)$

DSA Concepts Used

- Arrays
- Matrix/Grid Representation
- Linear Search
- Boolean State Tracking

4.5 Ticket Booking Algorithm

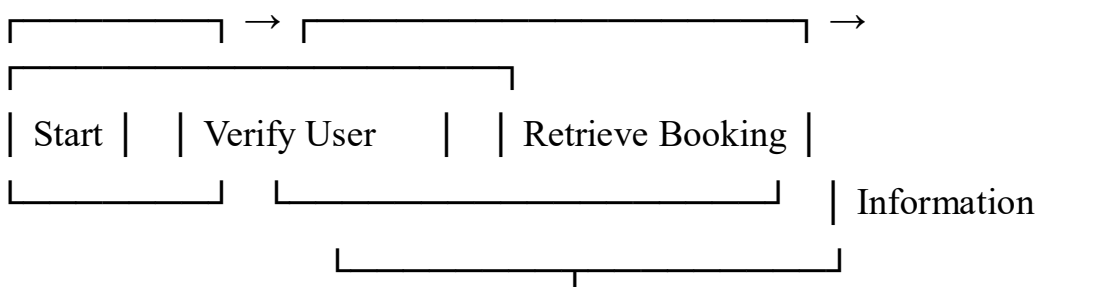
Objective

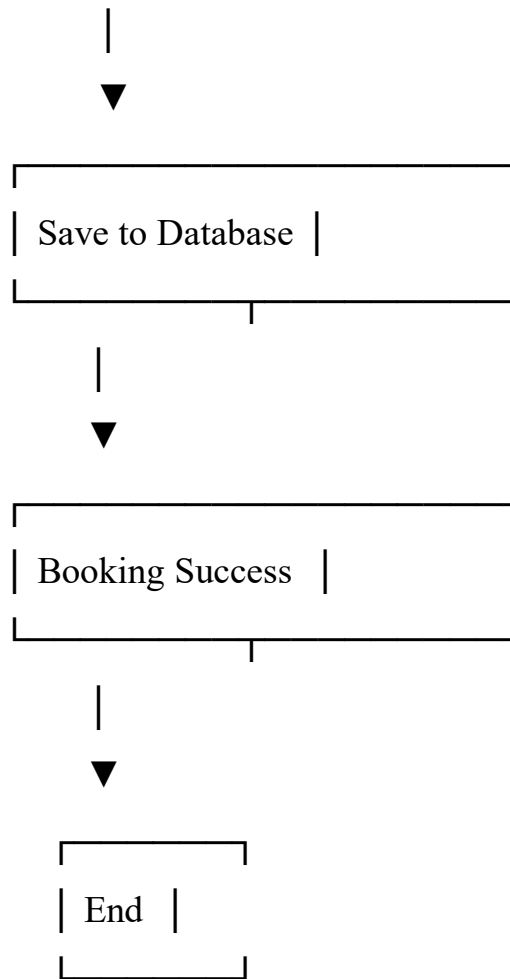
To confirm the booking and store ticket details in the database.

Algorithm Steps

1. Verify user authentication.
2. Retrieve selected movie.
3. Retrieve selected seats.
4. Generate booking details.
5. Save booking in the database.
6. Display booking confirmation.

Flowchart





Time Complexity $O(1)$

Space Complexity $O(1)$

DSA Concepts Used

- CRUD Operations
- Database Insertion
- Record Management

4.6 Booking History Retrieval Algorithm

Objective

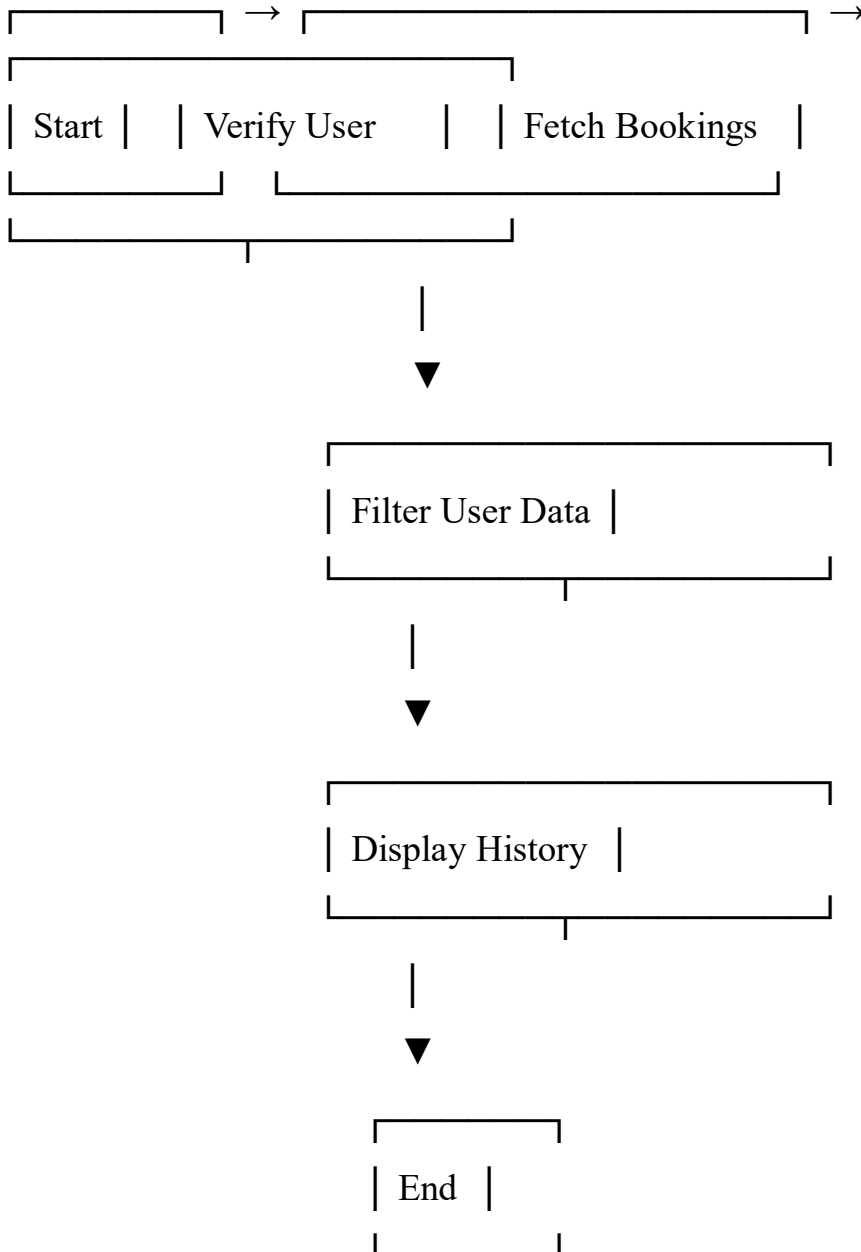
To display all previous bookings of the logged-in user.

Algorithm Steps

1. Verify user authentication.
2. Fetch booking records.

3. Filter bookings for the current user.
4. Display booking history.

Flowchart



Time Complexity $O(n)$

Space Complexity $O(n)$

DSA Concepts Used

- Arrays
- Filtering
- Traversal

- Database Query

Algorithm Summary

Algorithm	Time Complexity	Space Complexity
User Registration	$O(1)$	$O(1)$
User Login	$O(1)$	$O(1)$
Movie Fetching	$O(n)$	$O(n)$
Seat Selection	$O(n)$	$O(n)$
Ticket Booking	$O(1)$	$O(1)$
Booking History Retrieval	$O(n)$	$O(n)$

5. Results

The “**CineWave – Movie Ticket Booking System**” was successfully developed and tested using modern web technologies. The application performed as expected, providing users with a smooth, secure, and responsive movie ticket booking experience. All major functionalities, including user authentication, movie browsing, seat selection, ticket booking, and booking history, were tested and verified.

5.1 Home Page

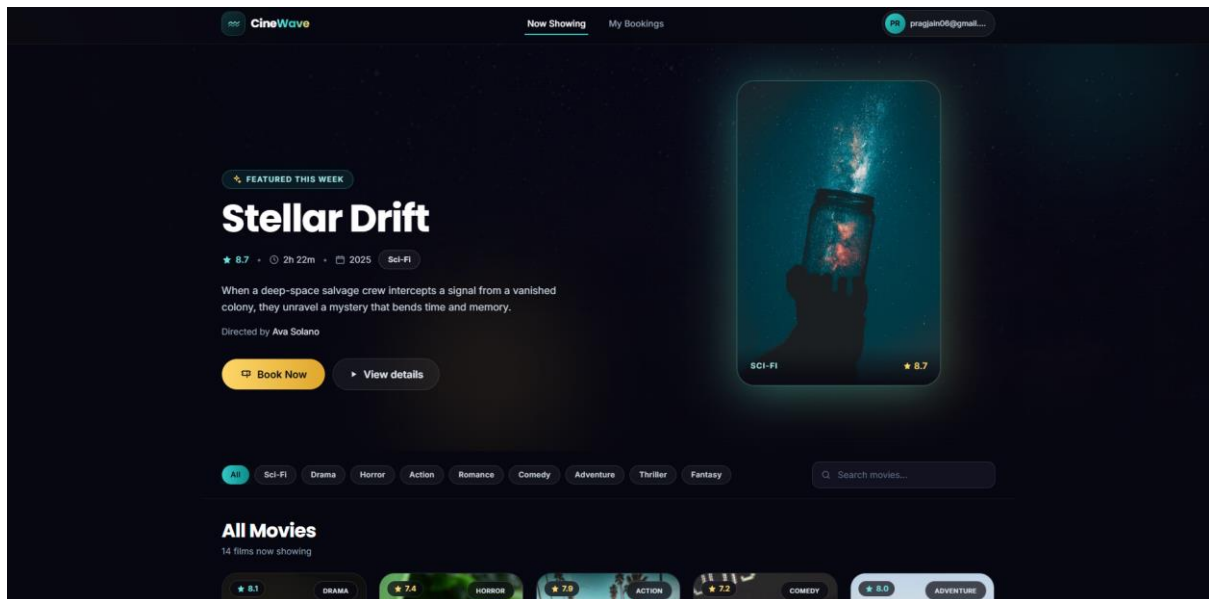
Description

The Home Page serves as the entry point of the application. It displays featured movies along with their posters and basic information in an organized and responsive layout.

Features

- Responsive navigation bar
- Featured movie section
- Movie cards with posters

- Clean and modern UI
- Mobile-friendly design



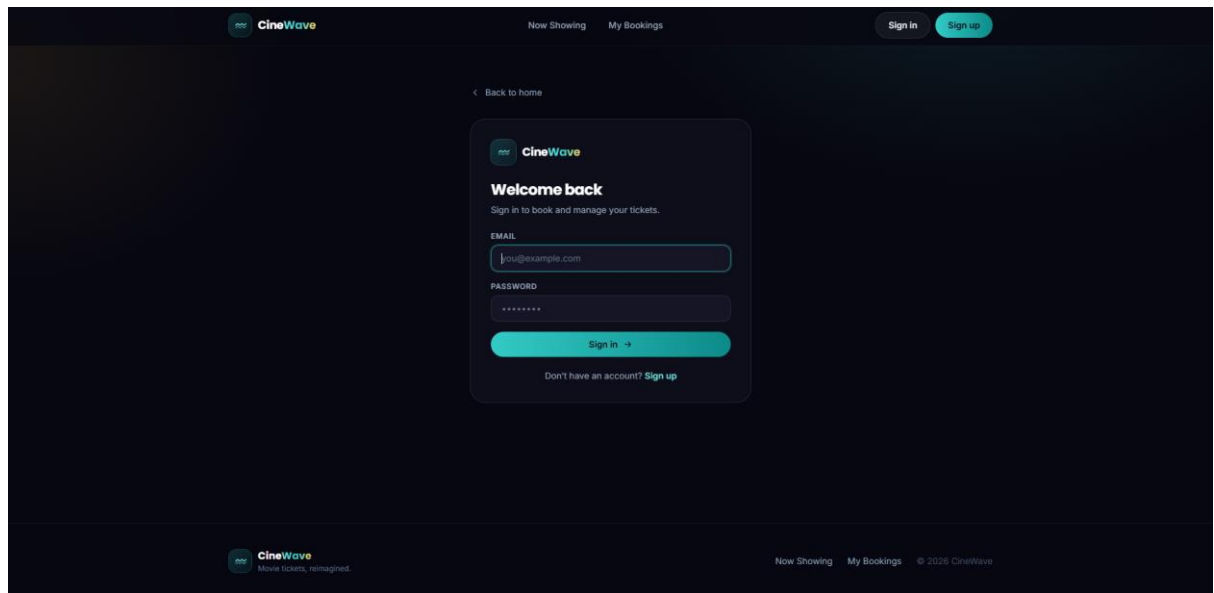
5.2 User Authentication

Description

The authentication module allows users to create a new account and securely log in using Supabase Authentication.

Features

- User Registration
- Secure Login
- Input Validation
- Session Management
- Error Handling



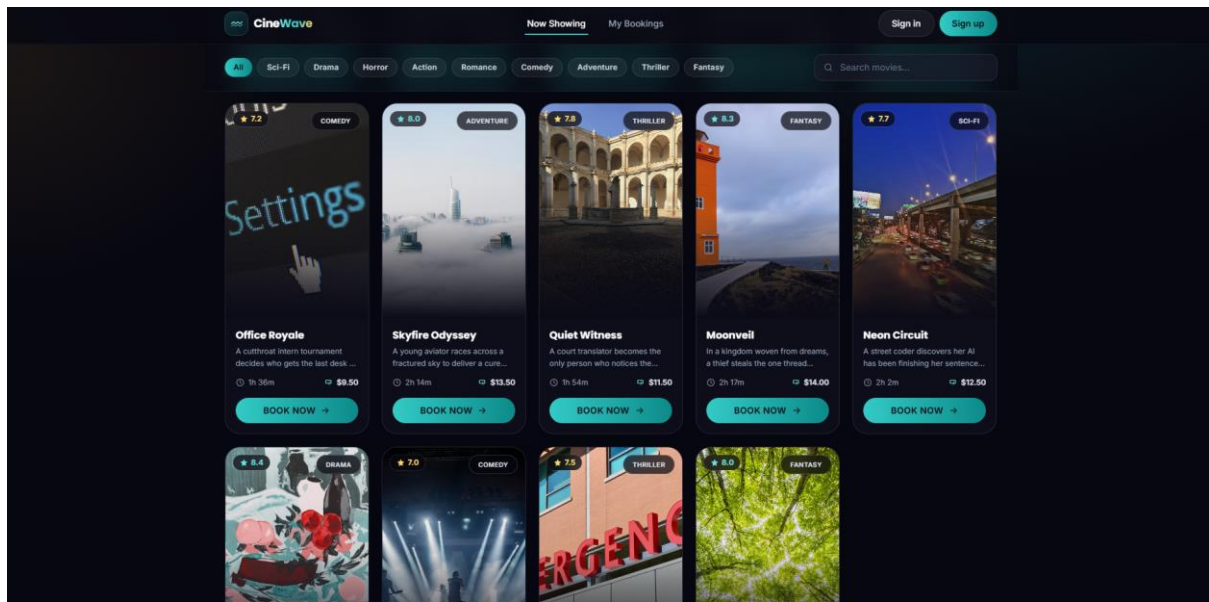
5.3 Movie Listing

Description

The Movie Listing page retrieves movie information from the PostgreSQL database and displays it dynamically using React components.

Features

- Dynamic movie fetching
- Movie posters
- Movie title and details
- Responsive card layout
- Real-time data rendering



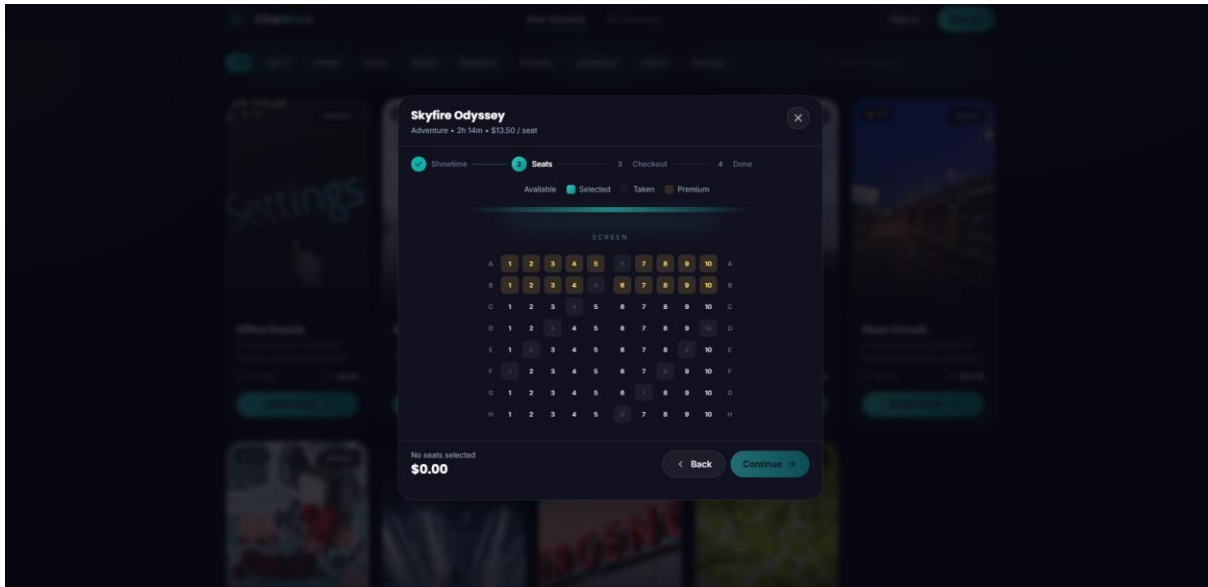
5.4 Seat Selection

Description

The Seat Selection module enables users to choose their preferred seats before confirming the booking. Available and booked seats are displayed separately to avoid duplicate reservations.

Features

- Interactive seat layout
- Available and booked seat indication
- Seat selection validation
- User-friendly interface



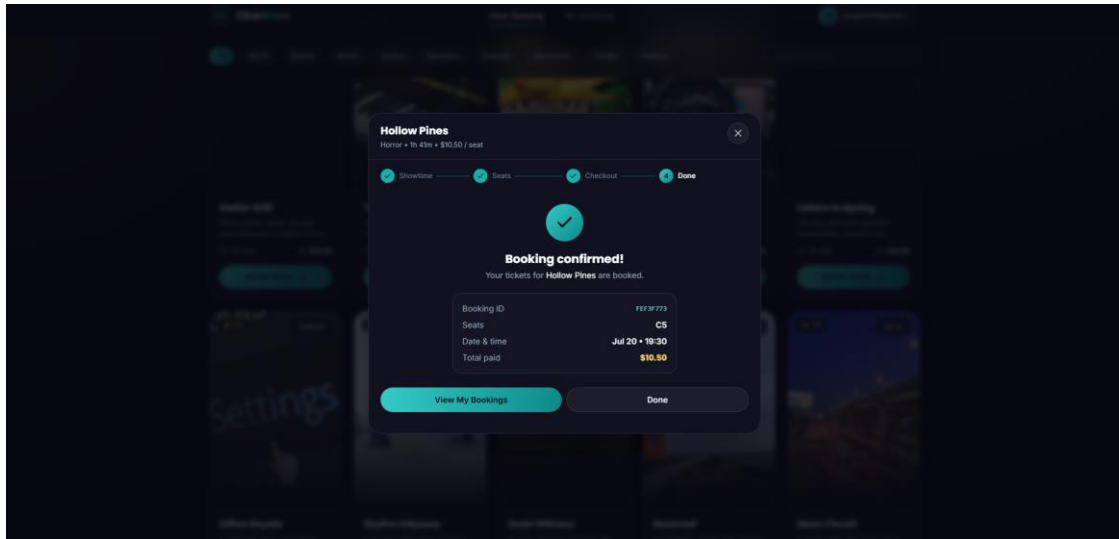
5.5 Ticket Booking

Description

After selecting the movie and seats, users can confirm their booking. The booking details are securely stored in the PostgreSQL database through Supabase.

Features

- Booking confirmation
- Secure database storage
- Automatic record generation
- User-specific bookings



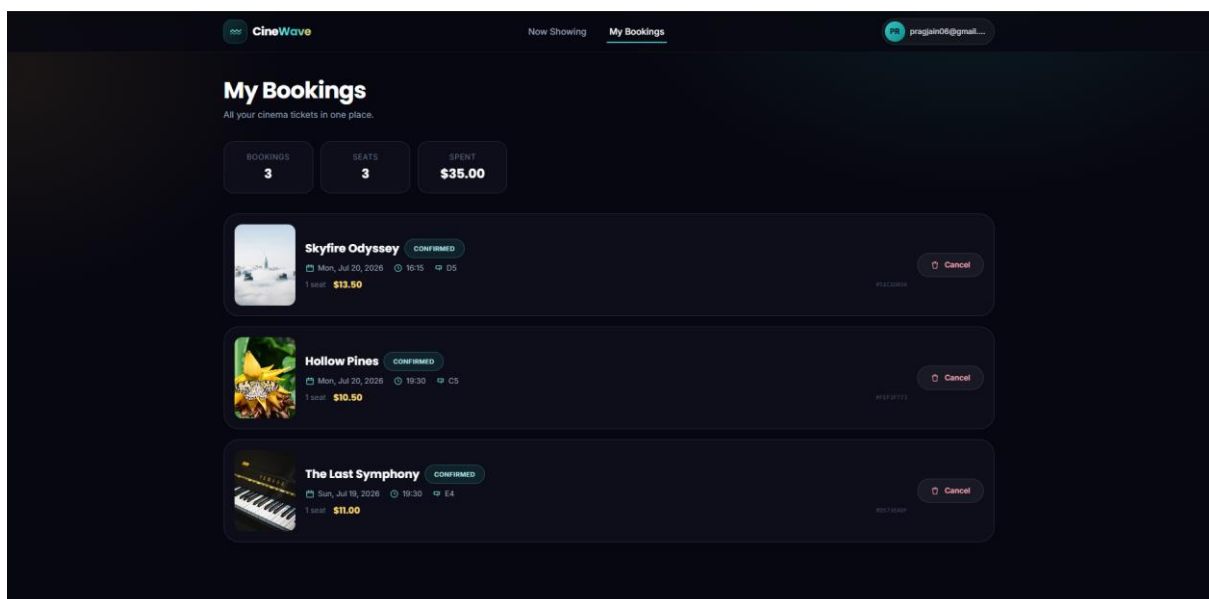
5.6 Booking History

Description

Users can view all their previous bookings through the Booking History page. Only authenticated users can access their personal booking records.

Features

- Booking history display
- User-specific records
- Secure data retrieval
- Responsive layout



5.7 Database Integration

Description

Supabase provides backend services, authentication, and PostgreSQL database management for the application.

Database Features

- PostgreSQL Database
 - Secure Authentication
 - Row Level Security (RLS)
 - CRUD Operations
 - Cloud Storage
 - Fast Data Retrieval
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5.8 Testing Results

- The application was tested for all major functionalities to ensure smooth operation and data consistency.

• Module	• Test Case	• Result
• User Registration	• New user registration	• Passed
• User Login	• Valid login credentials	• Passed
• Movie Fetching	• Retrieve movie list	• Passed
• Seat Selection	• Select available seats	• Passed
• Ticket Booking	• Store booking details	• Passed
• Booking History	• Retrieve user bookings	• Passed
• Database Connection	• PostgreSQL connection	• Passed
• Responsive Design	• Desktop & Mobile View	• Passed

6. Conclusion

The **CineWave – Movie Ticket Booking System** was successfully developed as a modern full-stack web application that simplifies the process of booking movie tickets online. The project combines a responsive frontend with a secure backend to provide users with a smooth and efficient booking experience.

The application enables users to register, log in securely, browse available movies, select preferred seats, book tickets, and view their booking history. By integrating **React.js**, **Tailwind CSS**, **Supabase**, and **PostgreSQL**, the system ensures reliable performance, secure data management, and an intuitive user interface.

The project also demonstrates the practical implementation of software engineering principles such as modular architecture, component-based development, client-server communication, cloud database integration, and user authentication. Additionally, it incorporates Data Structures and Algorithms (DSA) concepts including arrays, searching, filtering, and efficient database operations.

Objectives Achieved

- Developed a responsive movie ticket booking website.
- Implemented secure user authentication using Supabase.
- Created dynamic movie listing and detail pages.
- Successfully integrated interactive seat selection.
- Stored booking information securely in PostgreSQL.
- Displayed user-specific booking history.
- Successfully deployed the application on Vercel.

Advantages of the System

- Simple and user-friendly interface.
- Secure authentication and authorization.
- Fast and responsive performance.

- Cloud-based database management.
 - Modular and scalable architecture.
 - Accessible from desktop and mobile devices.
 - Easy maintenance and future enhancement.
-

Future Enhancements

The application can be further improved by adding the following features:

- Online payment gateway integration (Razorpay/Stripe).
- QR Code-based digital ticket generation.
- Email and SMS booking confirmation.
- Ticket cancellation and refund management.
- Movie ratings and user reviews.
- Admin dashboard for theatre and movie management.
- Multiple theatre and city support.
- Real-time seat synchronization.
- Recommendation system based on user preferences.
- AI-powered movie suggestions.

7. References

- The following resources were referred to during the design and development of the project:
- React Documentation. <https://react.dev/>
- Vite Documentation. <https://vitejs.dev/>
- Tailwind CSS Documentation. <https://tailwindcss.com/docs>
- Supabase Documentation. <https://supabase.com/docs>
- PostgreSQL Documentation. <https://www.postgresql.org/docs/>
- JavaScript Documentation (MDN Web Docs). <https://developer.mozilla.org/>

- Vercel Documentation. <https://vercel.com/docs>
- HTML5 Specification (W3C). <https://www.w3.org/TR/html5/>
- CSS Documentation (MDN Web Docs). <https://developer.mozilla.org/en-US/docs/Web/CSS>
- Node.js Documentation. <https://nodejs.org/docs/>